Institute for
Data Science Foundations
E-21

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Lecture 05

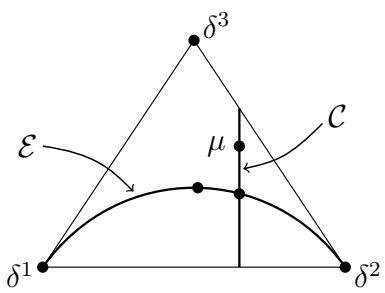
In the last lecture, we have introduced exponential families based on the maximum entropy method. Given a probability measure $\mu \in \mathcal{P}_+(I)$ and functions $f_1, ..., f_d$ on I, we have considered the convex set

$$\begin{aligned}\mathcal{C} &:= \mathcal{C}(\mu, f_1, ..., f_d) \\ &:= \{\nu \in \mathcal{P}_+(I) : \mathbb{E}_\nu(f_j) = \mathbb{E}_\mu(f_j), \quad j = 1, ..., d\}\end{aligned}$$

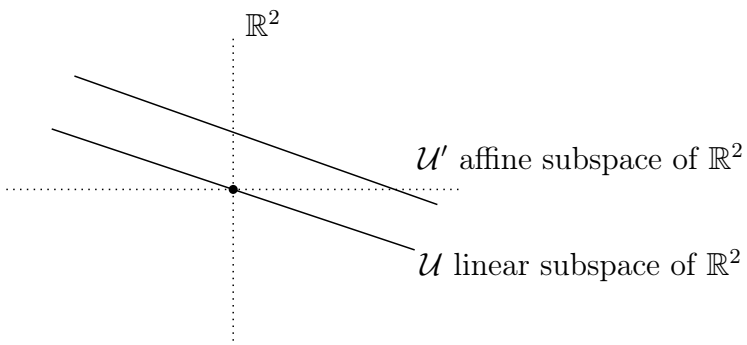
If we maximise the entropy in that set, we obtain a probability measure in the exponential family

$$\mathcal{E} := \mathcal{E}(f_1, ..., f_d) := \left\{ \frac{1}{Z(\theta)} \exp\left(\sum_{j=1}^{d} \theta_j f_j \right) : \theta_1, ..., \theta_d \in \mathbb{R} \right\}$$

Here is an illustration:



While $\mathcal{C}$ appears to be "flat", in some intuitive sense, $\mathcal{E}$ seems to be more complicated. However, this is not at all the case. It turns out that exponential families can be considered as linear or affine subspaces of $\mathcal{P}_+(I)$ in a way that we are now going to elaborate on. Let us first consider the situation in $\mathbb{R}^n$:



Consider a linear subspace $\mathcal{U}$ of $\mathbb{R}^n$. This is a flat geometric object which we can easily describe. It satisfies:

$$\left.\begin{array}{lrcl} (i) & v, w \in \mathcal{U} & \Rightarrow & v + w \in \mathcal{U} \\ (ii) & v \in \mathcal{U}, \lambda \in \mathbb{R} & \Rightarrow & \lambda v \in \mathcal{U} \end{array}\right\} \tag{1}$$

Now consider some $v_0 \in \mathbb{R}^n$ and the translation of $\mathcal{U}$ by v_0, that is

$$\mathcal{U}' := v_0 + \mathcal{U} = \{v_0 + v : v \in \mathcal{U}\}.$$

Clearly $\mathcal{U}'$ is as simple/flat as $\mathcal{U}$, but it does not satisfy the conditions (1) anymore. We have to find an alternative way to describe $\mathcal{U}'$. In addition to

$$p \in \mathcal{U}', v \in \mathcal{U} \Rightarrow p + v \in \mathcal{U}',$$

the following holds:

$$\begin{array}{ll}
(i) & p \in \mathcal{U}', \quad 0 \in \mathcal{U} \quad \Rightarrow p + 0 = p \\
(ii) & p \in \mathcal{U}', \quad v, w \in \mathcal{U} \quad \Rightarrow p + (v + w) = (p + v) + w \\
(iii) & \text{For all } p, q \in \mathcal{U}', \text{ there exists a unique } v \in \mathcal{U}, \text{such that } q = p + v.
\end{array}$$

More generally, an *affine space* E over a vector space V comes with a translation map

$$+ : E \times V \to E, \quad (p, v) \mapsto p + v.$$

Note that "+" can be very general, but the following conditions are required:

$$\begin{array}{ll}
(i) & p + 0 = p \\
(ii) & (p + v) + w = p + (v + w) \\
(iii) & \text{For all } p, q \in E, \text{ there exists a unique } v \in V, \text{such that } q = p + v.
\end{array}$$

We now define a translation on $\mathcal{M}_+(I)$:

$$\begin{aligned}
\mathcal{M}_+(I) \times \mathcal{F}(I) &\to \mathcal{M}_+(I) \\
(\mu, f) &\mapsto \mu + f := e^f \mu.
\end{aligned}$$

Obviously,

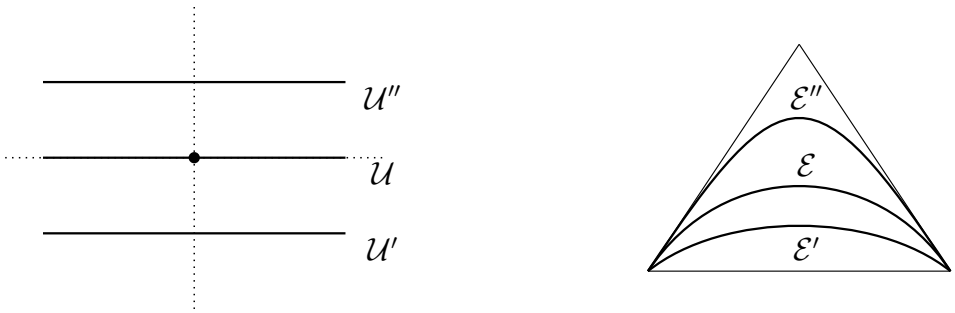
$$\begin{array}{ll}
(i) & \mu + 0 = e^0 \mu = 1 \cdot \mu = \mu \\
(ii) & (\mu + f) + g = e^g (e^f \mu) = (e^g e^f)\mu = e^{f+g}\mu = \mu + (f + g) \\
(iii) & \text{For } \mu, \nu \in \mathcal{M}_+(I), \text{ choose} \\
& \qquad f = \ln\left(\dfrac{d\nu}{d\mu}\right) \\
& \text{Then, } \mu + f = \mu + \ln \dfrac{d\nu}{d\mu} \\
& \qquad = e^{\ln \frac{d\nu}{d\mu}} \mu = \dfrac{d\nu}{d\mu}\mu = \nu \\
& \text{(Note that uniqueness of } f \text{ remains to be shown.)}
\end{array}$$

We use this to define a translation on $\mathcal{P}_+(I)$:

$$\begin{aligned}
\mathcal{P}_+(I) \times (\mathcal{F}(I)/\mathbb{R}) &\to \mathcal{P}_+(I) \\
(\mu, f + \mathbb{R}) &\mapsto \mu + (f + \mathbb{R}) := \frac{e^f}{\mathbb{E}_\mu(e^f)}\mu.
\end{aligned}$$

One can easily verify that the properties $(i), (ii)$, and $(iiii)$ also hold in this case. (Exercise)

One can easily see that the affine subspaces of $\mathcal{P}_+(I)$ are exactly the exponential families, and they even correspond to affine subspaces of $\mathbb{R}^n$:



Thus, exponential families are simple/flat objects. We will see that the maximum entropy estimation as well as the maximum likelihood estimation can be interpreted as orthogonal projection on an exponential family. Let us start with the maximum entropy method and consider the exponential family $\mathcal{E}$ parametrised by

$$\varphi : \theta = (\theta_1, ..., \theta_d) \mapsto \frac{e^{\sum_{j=1}^d \theta_j f_j}}{\sum_{i \in I} e^{\sum_{j=1}^d \theta_j f_j(i)}} = \frac{e^{\sum_{j=1}^d \theta_j f_j}}{Z(\theta)}$$

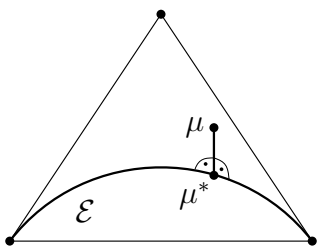
In order to determine the tangent space of $\mathcal{E}$ in $\varphi(\theta)$, we have to take the partial derivatives:

$$\begin{aligned}
\frac{\partial \varphi}{\partial \theta_r}(\theta) &= \frac{e^{\sum_{j=1}^d \theta_j f_j(i)} \cdot f_r \cdot Z(\theta) - \left(\sum_{i \in I} e^{\sum_{j=1}^d \theta_j f_j(i)} f_r(i)\right) \cdot e^{\sum_{j=1}^d \theta_j f_j}}{Z^2(\theta)} \\
&= \frac{e^{\sum_{j=1}^d \theta_j f_j}}{Z(\theta)} f_r \; - \; \left(\sum_{i \in I} \frac{e^{\sum_{j=1}^d \theta_j f_j(i)}}{Z(\theta)} f_r(i)\right) \cdot \frac{e^{\sum_{j=1}^d \theta_j f_j}}{Z(\theta)} \\
&= \varphi(\theta) \cdot f_r - \mathbb{E}_{\varphi(\theta)}(f_r) \; \varphi(\theta) \\
&= \varphi(\theta) \left(f_r - \mathbb{E}_{\varphi(\theta)}(f_r)\right), \quad r = 1, ..., d
\end{aligned}$$

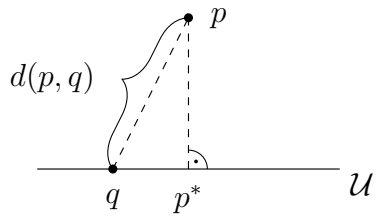
We now consider the difference vector $\mu - \mu^*$ between the initial probability measure $\mu \in \mathcal{P}_+(I)$ and its maximum entropy estimate μ^*, which is an element of $\mathcal{E}$ so that there is a θ^* satisfying $\mu^* = \varphi(\theta^*)$. For the Fisher-Rao inner products we obtain

$$\begin{aligned}
\left\langle \mu - \mu^*, \frac{\partial \varphi}{\partial \theta_r}(\theta^*) \right\rangle_{\varphi(\theta^*)} &= \sum_{i \in I} \frac{1}{\mu_i^*} \; \varphi_i(\theta^*) \; \left(f_r(i) - \mathbb{E}_{\varphi(\theta^*)}(f_r)\right) \; (\mu_i - \mu_i^*) \\
&= \sum_{i \in I} \mu_i f_r(i) - \sum_{i \in I} \mu_i^* f_r(i) - \sum_{i \in I} \mathbb{E}_{\varphi(\theta^*)}(f_r)(\mu_i - \mu_i^*) \\
&= \mathbb{E}_\mu(f_r) - \mathbb{E}_{\mu^*}(f_r) = 0.
\end{aligned}$$

Thus, μ^* is an orthogonal projection, orthogonality with respect to the Fisher-Rao inner product.



In $\mathbb{R}^n$, orthogonal projections can also be characterised in terms of distance minimisation:



$$\iff \quad \begin{aligned} & p - p^* \perp \mathcal{U} \\ & d(p,p^*) \\ & = \min_{q \in \mathcal{U}} d(p,q) \end{aligned}$$

This works for the distance

$$d(p,q) = \sqrt{\sum_{i \in I} (p_i - q_i)^2}.$$

In order to translate this geometric picture to the setting of exponential families, we have to introduce a kind of distance function on $\mathcal{P}_+(I)$. We already studied various distance functions on $\mathcal{P}_+(I)$, the Hellinger distance, the Fisher-Rao distance, etc. However, in order to obtain the equivalence of orthogonal projection to distance minimisation, as in $\mathbb{R}^n$, we have to introduce another distance function which resembles the Shannon entropy.

Definition. The *Kullback-Leibler divergence (KL-divergence)* for $\mu, \nu \in \mathcal{P}(I)$ is defined as

$$D_{KL}(\mu \| \nu) = \sum_{i \in I} \mu_i \ln \frac{\mu_i}{\nu_i}.$$

Here, we use the convention

$$\mu_i \ln \frac{\mu_i}{\nu_i} = 0 \quad \text{whenever} \quad \mu_i = 0, \text{ and}$$

$$\mu_i \ln \frac{\mu_i}{\nu_i} = \infty \quad \text{whenever} \quad \mu_i > \nu_i = 0.$$

$\Diamond$

Proposition.

1. *$D_{KL}(\mu \| \nu) \geq 0$, and $D_{KL}(\mu \| \nu) = 0$ if and only if $\mu = \nu$.*
2. *D_{KL} is jointly convex, that is, for $\mu^{(j)}, \nu^{(j)}, \lambda_j \in [0,1]$, $j = 1, ..., n$, satisfying $\sum_{j=1}^n \lambda_j = 1$,*

$$D_{KL}\left(\sum_{j=1}^n \lambda_j \mu^{(j)} \parallel \sum_{j=1}^n \lambda_j \nu^{(j)} \right) \leq \sum_{j=1}^n \lambda_j D_{KL}\left(\mu^{(j)} \| \nu^{(j)}\right).$$

$\square$

We can now use the KL-divergence to project on an exponential family $\mathcal{E}$. The following nice theorem holds, which mimics the situation in $\mathbb{R}^n$:

Theorem. *Let $\mathcal{E}$ be an exponential family in $\mathcal{P}_+(I)$, and let $\mu \in \mathcal{P}_+(I)$. Then the following two conditions for a $\mu^* \in \mathcal{E}$ are equivalent*
(i) $\mu - \mu^* \perp X$ for all tangent vectors in $T_{\mu^*}\mathcal{E}$.
(ii) $D_{KL}(\mu \| \mu^*) = \min_{\nu \in \mathcal{E}} D_{KL}(\mu \| \nu)$.
Furthermore, there exists exactly one $\mu^* \in \mathcal{E}$ which satisfies one and therefore both conditions.

Note that the existence and uniqueness of a $\mu^* \in \bar{\mathcal{E}}$ which satisfies

$$D_{KL}(\mu \| \mu^*) = \min_{\nu \in \bar{\mathcal{E}}} D_{KL}(\mu \| \nu)$$

can also be proven if $\mu \in \mathcal{P}(I)$, not necessarily positive. Here, we have to replace the exponential family by its closure $\bar{\mathcal{E}}$. This allows us to also project an empirical distribution given by the samples $x_1, ..., x_m$:

$$\mu = \frac{1}{m} \sum_{i=1}^{m} \delta^{x_i}$$

Then

$$D_{KL}(\mu \| \mu_\theta) = \ln \frac{1}{m} - \frac{1}{m} \sum_{i=1}^{m} \ln \left(\mu(x_i; \theta) \right)$$

Minimising this is equivalent to maximising $\sum_{i=1}^{m} \ln \left(\mu(x_i; \theta) \right)$. This is nothing but the maximum likelihood method.